TASK 2: WINDOWS EVENT LOG ANALYSIS

SOC Analyst Internship | Security Event Investigation

1. Objective

The objective of this task was to review Windows Security events, identify successful and failed logons, inspect events involving special privileges and process creation, and look for evidence that could warrant further investigation of brute-force activity or privilege escalation.

2. Tools and Methodology

Windows Event Viewer was used to open Windows Logs > Security. The log was filtered by selected Event IDs and the event counts and General-tab details were reviewed. Screenshots were captured as supporting evidence.

1. Opened Event Viewer (eventvwr.msc) and selected Windows Logs > Security.
2. Filtered the Security log by Event ID 4625 and reviewed failed-logon timestamps and one event's details.
3. Filtered by Event ID 4624 and inspected a successful service logon.
4. Filtered by Event ID 4672 to inspect special privileges assigned to a logon session.
5. Filtered by Event ID 4688 to inspect a process-creation event.
6. Filtered by Event ID 1102 to check for recorded Security audit-log clearing events.

3. Event Summary

Event ID	Meaning	Observed count
4624	Successful logon	5,358
4625	Failed logon	20
4672	Special privileges assigned to a new logon	5,103
4688	A new process was created	180
1102	Security audit log was cleared	0

Note: These counts are the values displayed by Event Viewer for the filters during this review. The count of 4625 events was displayed as 20; 17 timestamps were transcribed during the review.

4. Detailed Findings

4.1 Event ID 4625 — Failed Logons

Event ID 4625 records a failed logon attempt. The reviewed event was logged on 22 October 2026 at 6:58:11 PM. It showed Logon Type 2 (interactive), the failure reason “Unknown user name or bad password,” status 0xC000006D, and substatus 0xC000006A, which indicates an incorrect password. The process shown was Microsoft Edge WebView2 (msedgewebview2.exe). The source network address was blank.

The recorded failed-logon timestamps were spread across multiple dates from 17 September to 22 October 2026, with a few events occurring relatively close together. The supplied timestamps alone do not establish a sustained brute-force attack. The failures could have several explanations, including mistyped credentials or an application request; further context would be needed.

4.2 Event ID 4624 — Successful Logons

Event Viewer displayed 5,358 events for Event ID 4624. The inspected event had Logon Type 5, which represents a service logon. The new logon account was NT AUTHORITY\SYSTEM, and the process was C:\Windows\System32\services.exe. The Elevated Token field was Yes. A SYSTEM service logon with elevated privileges can be normal Windows behavior and, by itself, is not evidence of privilege escalation.

4.3 Event ID 4672 — Special Privileges Assigned

Event Viewer displayed 5,103 events for Event ID 4672. The inspected event belonged to NT AUTHORITY\SYSTEM with Logon ID 0x3E7. Listed privileges included SeDebugPrivilege, SeBackupPrivilege, SeRestorePrivilege, SeTakeOwnershipPrivilege, and SeImpersonatePrivilege. These are powerful privileges, but their presence in a SYSTEM session is commonly expected. This event should be correlated with the related logon and process events before treating it as suspicious.

4.4 Event ID 4688 — Process Creation

Event Viewer displayed 180 events for Event ID 4688. The reviewed event showed C:\Windows\System32\services.exe as the new process and C:\Windows\System32\wininit.exe as the creator process. The creator subject was SYSTEM and the mandatory label was System Mandatory Level. This is consistent with normal Windows service initialization. The process command line was blank in the event, so the event alone provides limited context for deeper investigation.

4.5 Event ID 1102 — Security Audit Log Cleared

The filter returned 0 events for Event ID 1102. No Security audit-log-clearing event was found in the filtered log during this review. This does not prove that logs have never been cleared; it only describes the available recorded events in the reviewed log.

5. AI-Assisted Security Interpretation

AI-assisted interpretation was used to explain the Windows Event IDs and distinguish common system activity from potential indicators that require correlation. The review did not identify enough evidence to confirm a brute-force attack or privilege-escalation incident.

A typical privilege-escalation investigation may look for a correlated sequence such as:

7. An unusual successful logon (4624), possibly preceded by repeated failed logons (4625).
8. A special-privileges event (4672) associated with the same account or logon session.
9. Suspicious process creation (4688), particularly an unexpected executable, unusual parent process, or suspicious command line.
10. A service installation or configuration change (for example, System event 7045 or Security event 4697, if available and auditing is enabled).
11. Unexpected account or group membership changes (for example, 4720, 4728, or 4732), followed by activity that uses the new privileges.

This is an investigative pattern, not a guaranteed sequence. Event IDs must be correlated by timestamp, account, logon ID, process path, command line, and host context. Events 4672 and 4688 can occur during normal Windows operation.

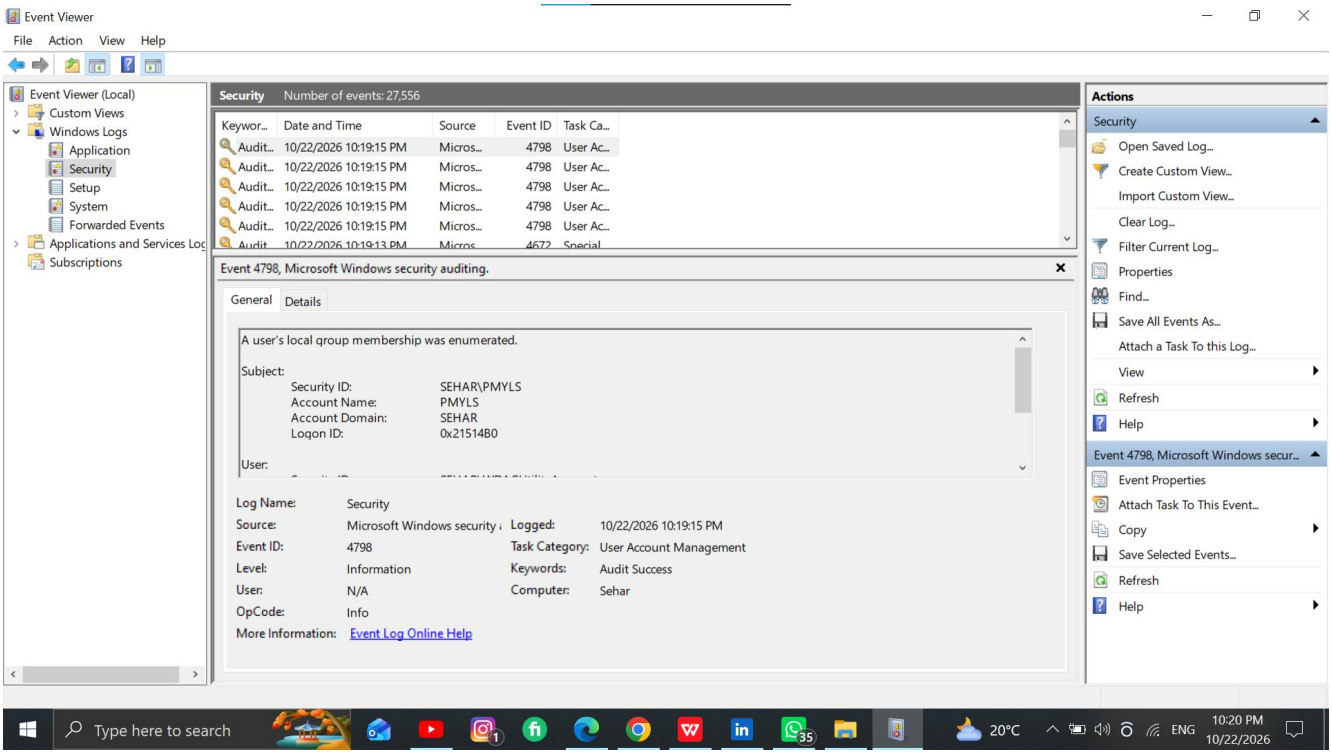
6. Conclusion

The Windows Security log review identified failed logon attempts, successful service logons, privileged SYSTEM sessions, and process-creation activity. The reviewed 4625 event indicated an incorrect password, while the examined 4624, 4672, and 4688 events were consistent with routine Windows system activity. No Event ID 1102 records were found in the filter. Based on the evidence reviewed, neither a brute-force attack nor privilege escalation can be confirmed. The findings are preliminary and limited to the selected events and available audit data.

7. Evidence / Screenshot Index

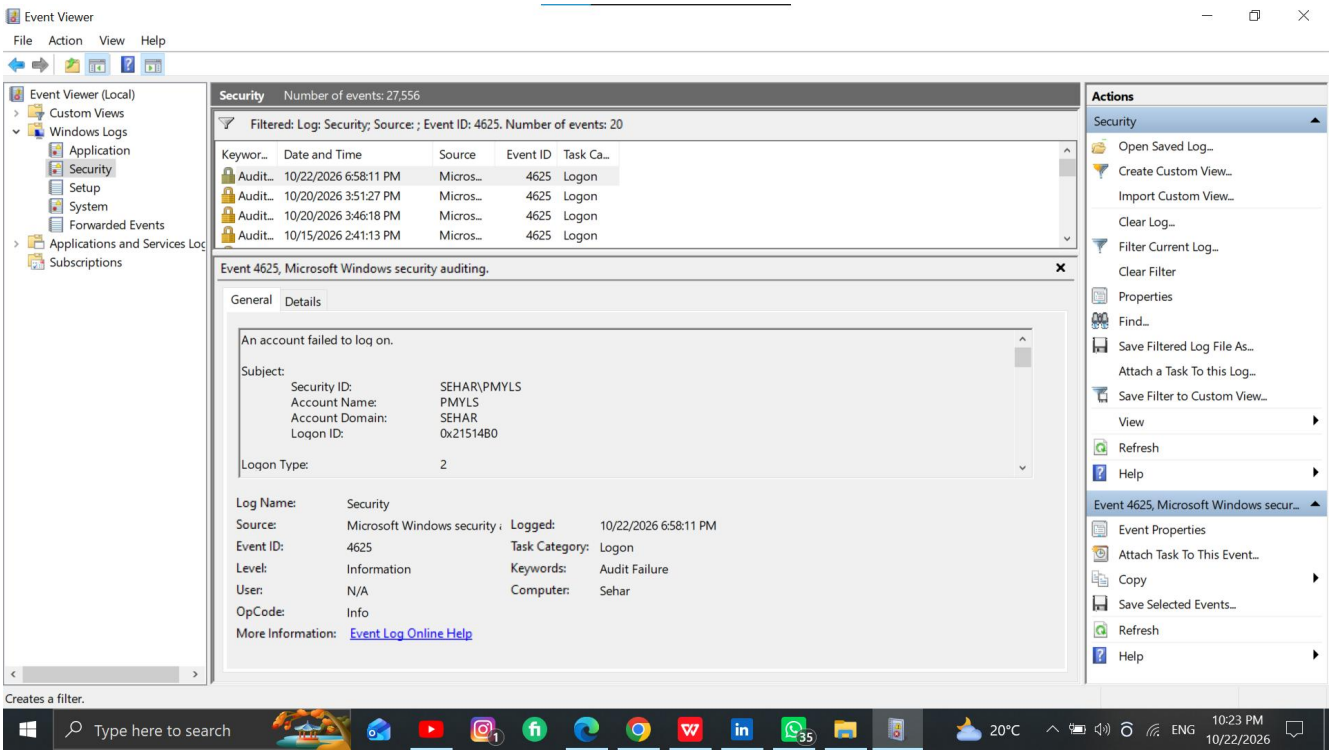
Screenshot 1: Windows Security Log Overview

Description: Overview of Windows Security logs in Event Viewer.



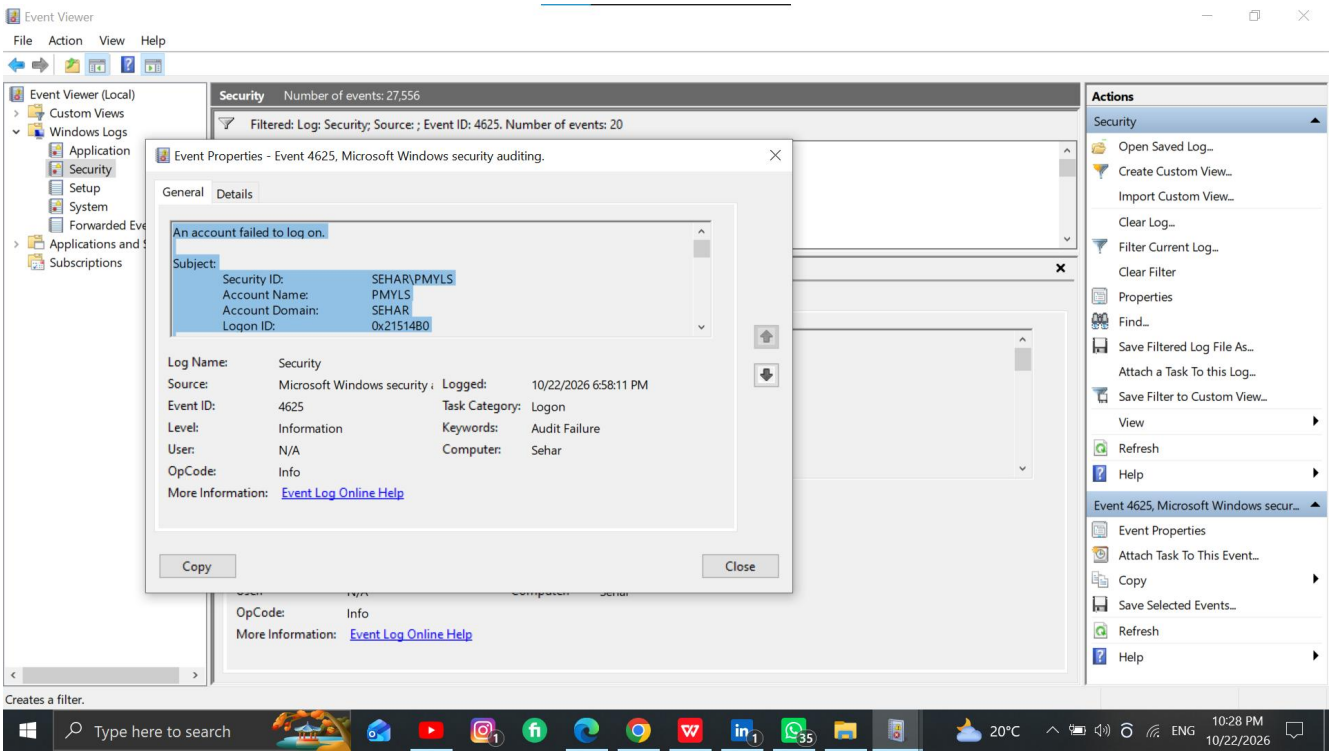
Screenshot 2: Failed Logon Events (Event ID 4625)

Description: Filtered Security log showing failed logon attempts.



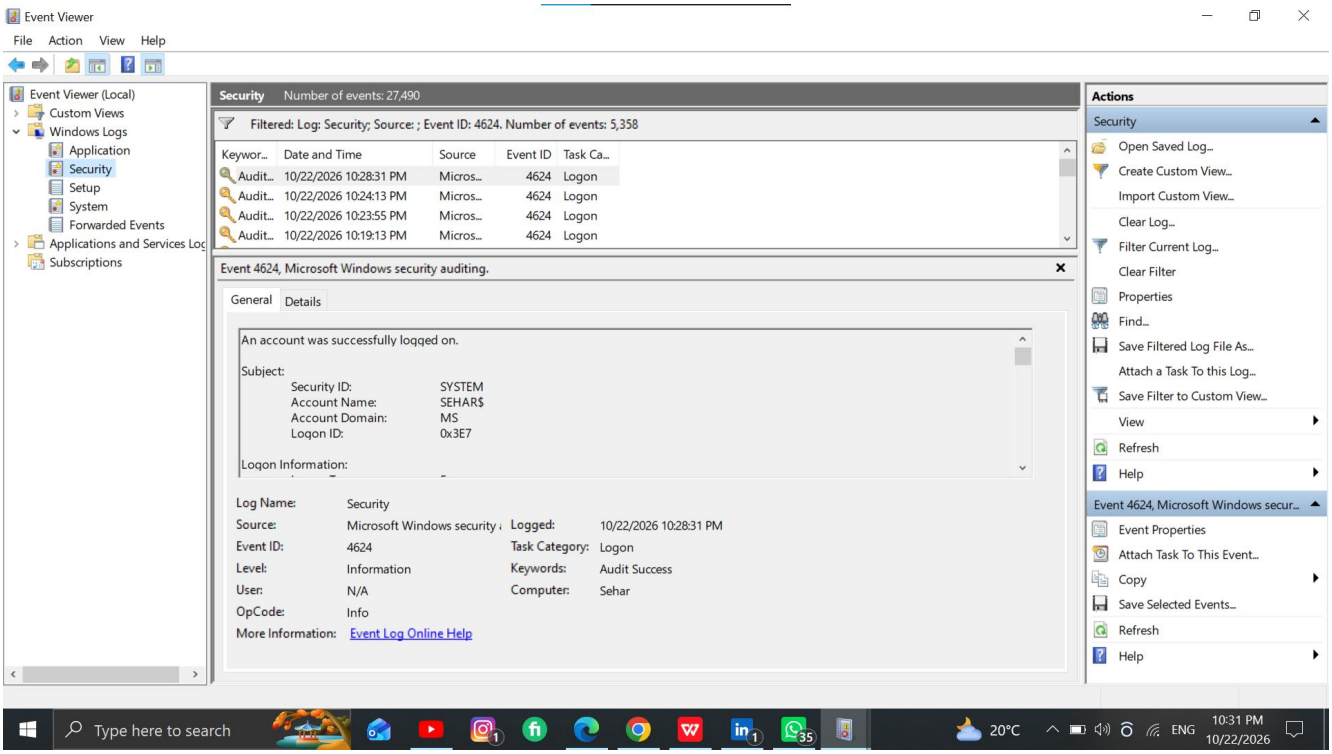
Screenshot 3: Failed Logon Details

Description: Details of a failed logon, including the failure reason and status codes.



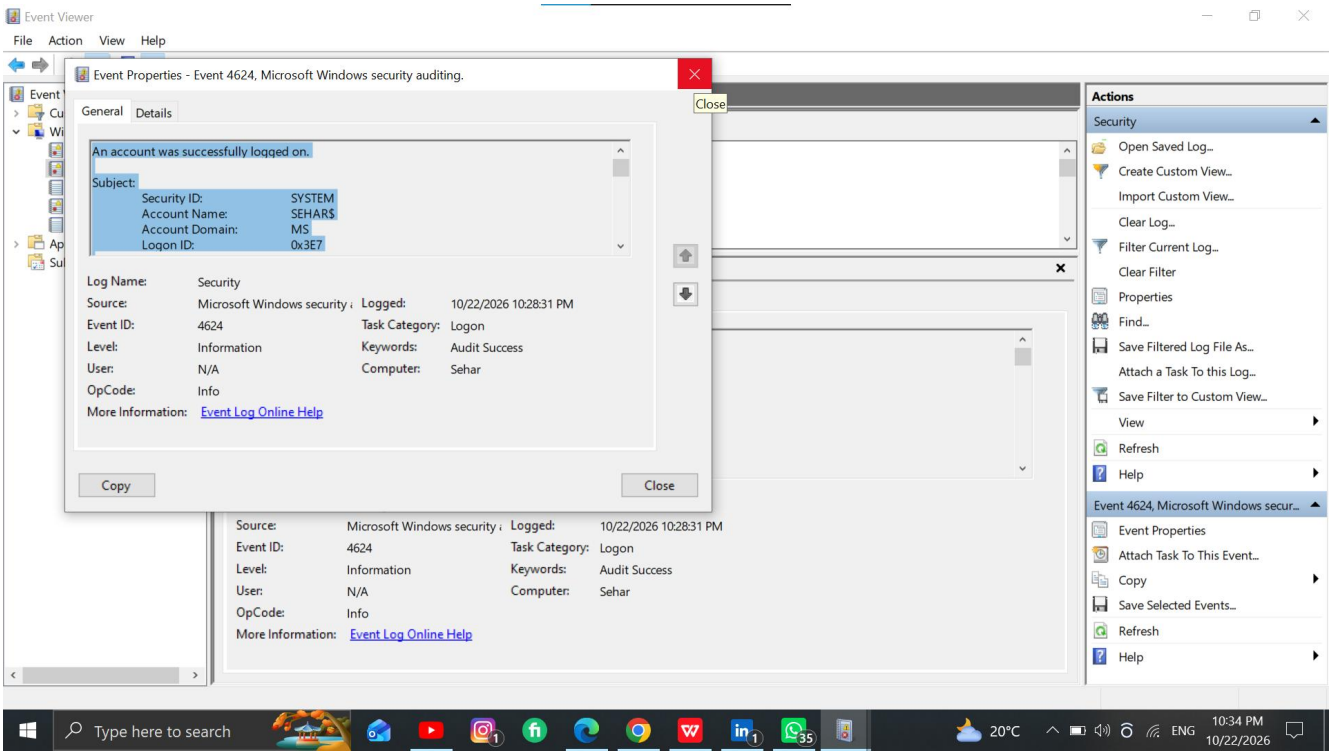
Screenshot 4: Successful Logon Events (Event ID 4624)

Description: Filtered list of successful logon events.



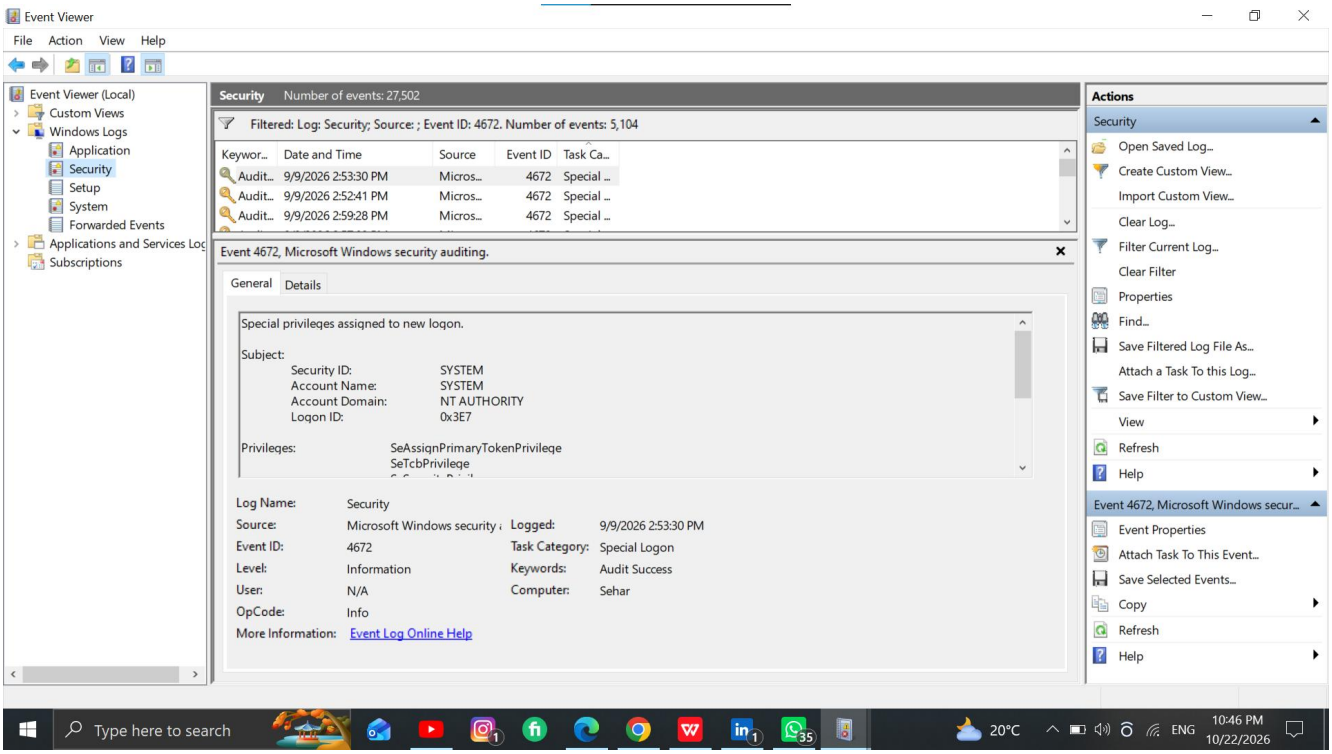
Screenshot 5: Successful Logon Details

Description: Details of a Windows service logon under the SYSTEM account.



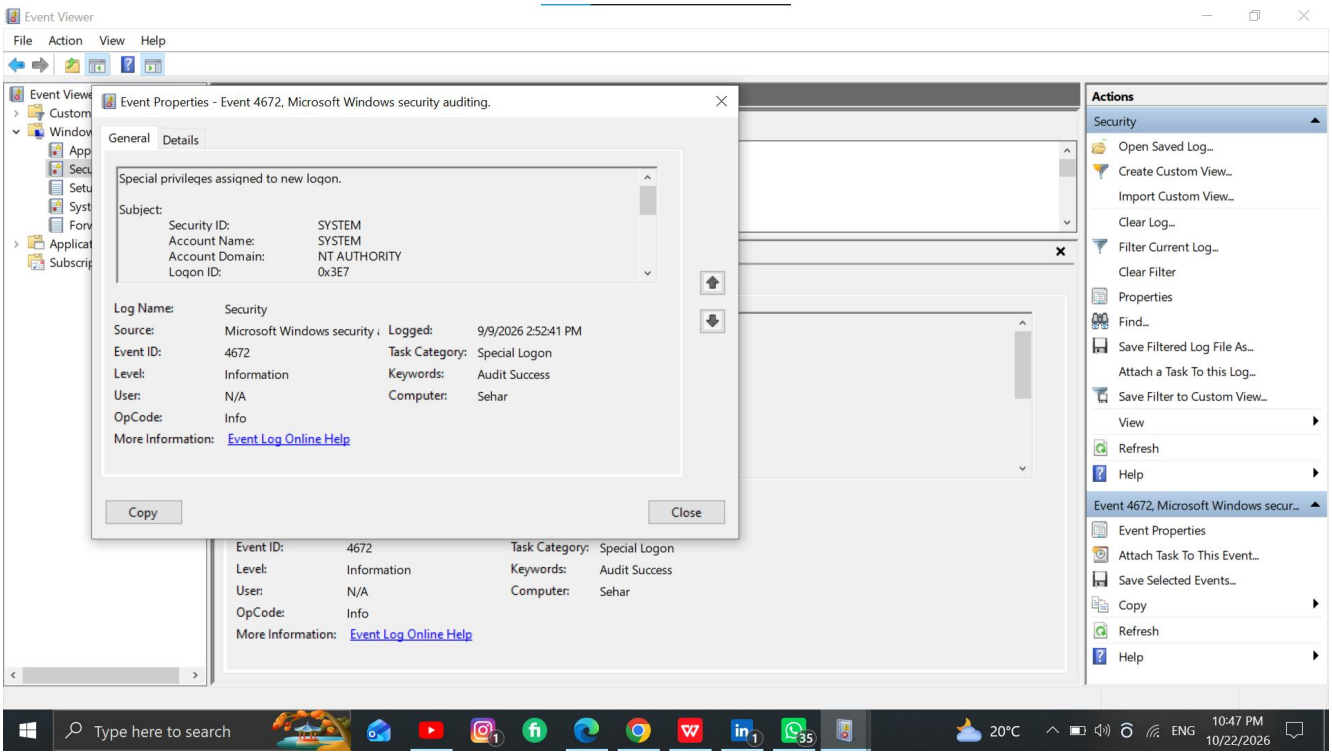
Screenshot 6: Special Privileges Events (Event ID 4672)

Description: Filtered list of events involving special privileges.



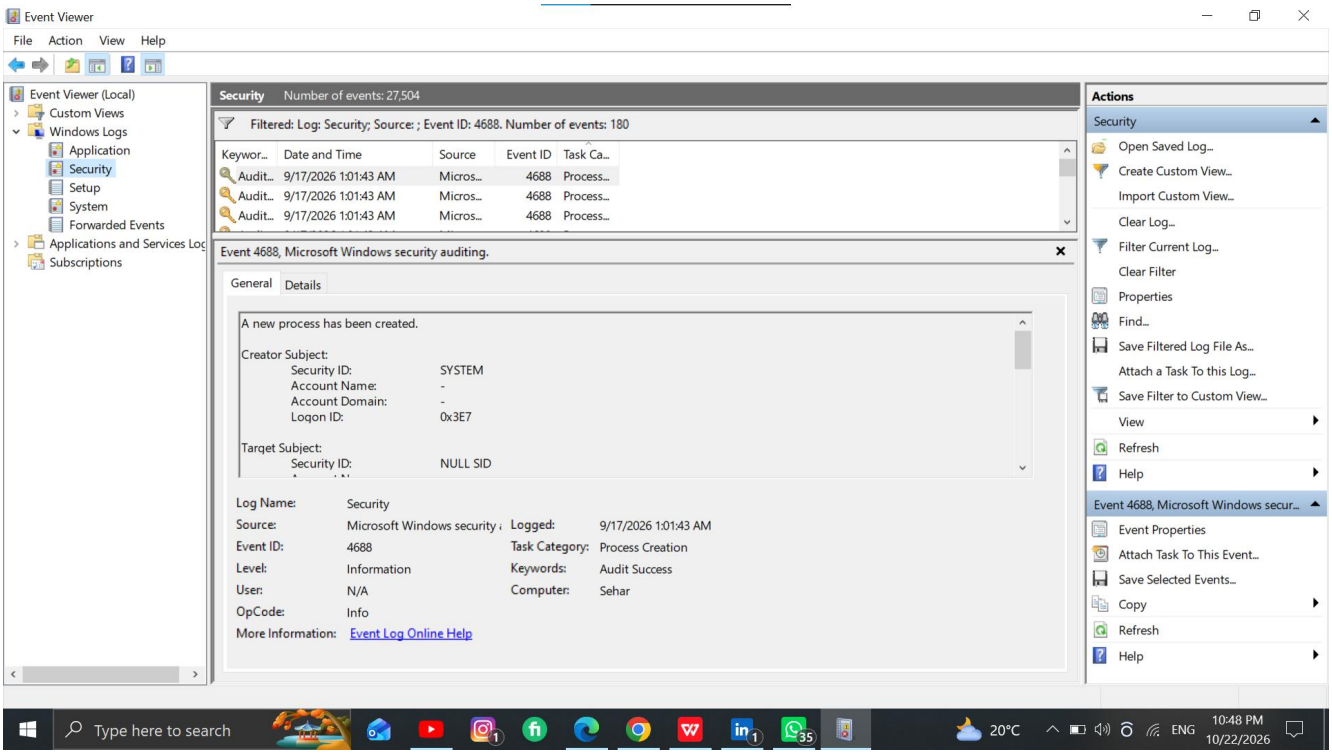
Screenshot 7: Special Privileges Details

Description: Details of privileges assigned to the SYSTEM account.



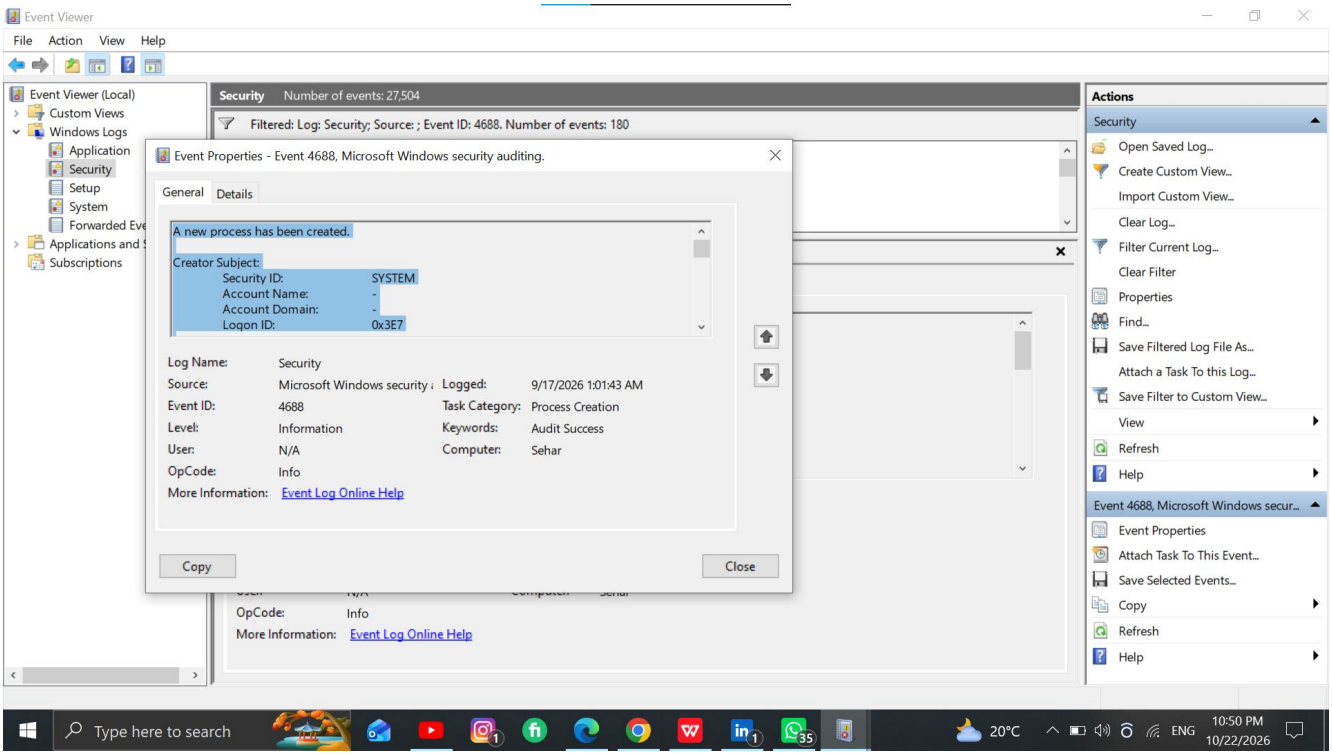
Screenshot 8: Process Creation Events (Event ID 4688)

Description: Filtered list of process-creation events.



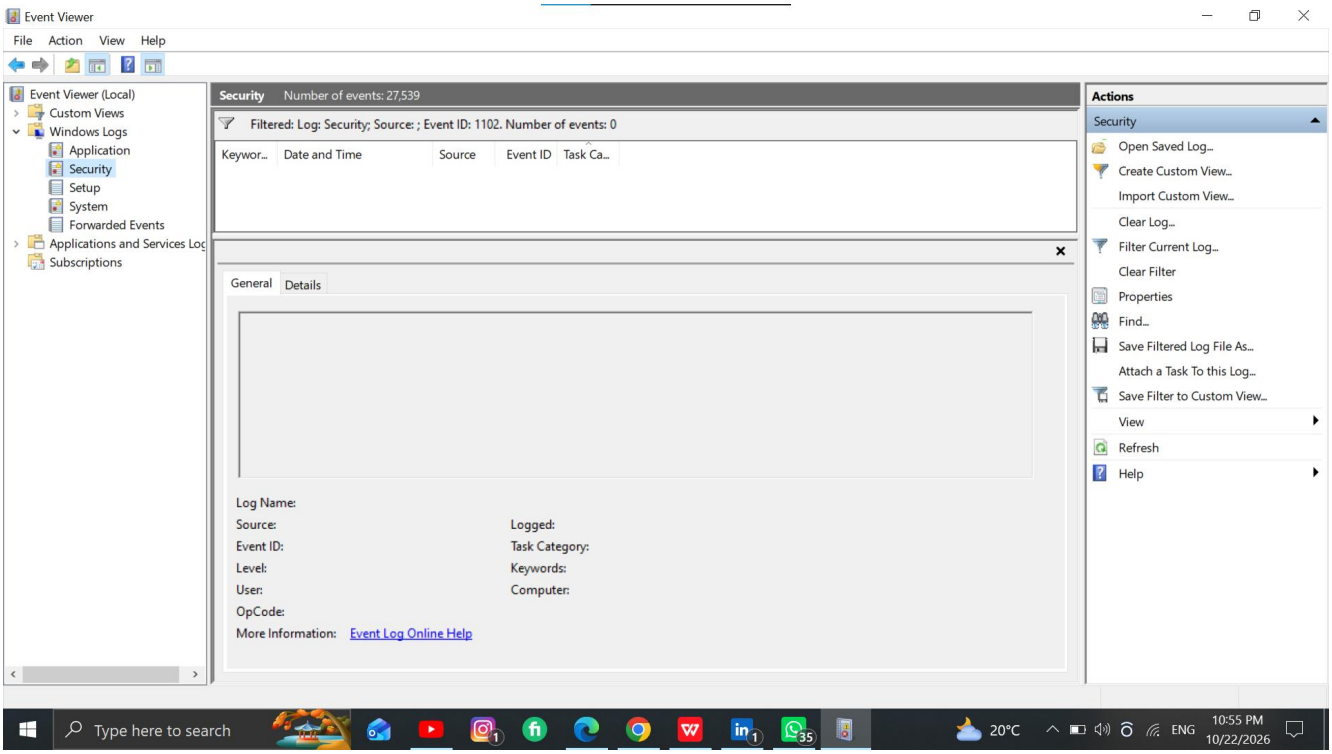
Screenshot 9: Process Creation Details

Description: Details showing the creation of services.exe by wininit.exe.



Screenshot 10: Security Audit Log Check (Event ID 1102)

Description: Filter result showing zero recorded events for Security audit-log clearing.



8. Limitations

- The analysis used selected events and displayed counts rather than a full forensic export of all records.
- Only one representative event's details were reviewed for Event IDs 4624, 4672, and 4688.
- The 4625 timestamps transcribed during the review included 17 entries, although Event Viewer displayed a count of 20.
- Blank source addresses and command-line fields limit attribution and process-level interpretation.
- Auditing settings, retention limits, and the completeness of the available Security log were not independently verified.

End of Report